

# NATIONAL TELE-OPHTHALMOLOGY SCREENING NETWORK

Automated Diabetic Retinopathy Diagnostic Validation Report — Rural PHC Deployment

## 1. PATIENT DEMOGRAPHICS & CLINICAL PROFILE

|                   |                   |                        |                       |
|-------------------|-------------------|------------------------|-----------------------|
| Patient Name:     | Anonymous Patient | Screening Session ID:  | e239c8a1-8ef2-4d33... |
| Age / Gender:     | 50 yrs / Female   | Contact Phone:         | +91 9876543210        |
| Diabetes History: | Type 2 (5 yrs)    | Screening Date & Time: | 2026-08-31 10:39:16   |

## 2. AI MULTI-MODAL DIAGNOSTIC FINDINGS (ICDR CRITERIA)

|                                                                           |                                                  |                        |       |                |      |
|---------------------------------------------------------------------------|--------------------------------------------------|------------------------|-------|----------------|------|
| ICDR Diagnosis:                                                           | Moderate NPDR (Grade 2 - Exudates & Hemorrhages) | Calibrated Confidence: | 97.4% | Image Quality: | GOOD |
| Recommended Clinical Directive: Ophthalmologist Referral within 4-6 Weeks |                                                  |                        |       |                |      |

## 3. QUANTITATIVE LESION & VASCULAR BIOMARKERS

| Biomarker Structure              | Count / Value | Clinical Pathology (ICDR Classification)            |
|----------------------------------|---------------|-----------------------------------------------------|
| ■ Microaneurysms / Hemorrhages   | 3             | Focal vascular outpouchings (Stage 1+ indicator)    |
| ■ Hard Exudates (Lipid Residue)  | 1             | Lipoprotein leakage marker (Moderate NPDR Stage 2+) |
| ■ Cotton Wool Spots (Ischemia)   | 0             | Axoplasmic flow blockage (Severe NPDR Stage 3+)     |
| ■ Retinal Vessel Network Density | 0.34%         | Normal reference baseline: 8.0% - 18.0%             |
| ■ Optic Disc Landmark Coordinate | (105, 45)     | Verified reference center for NVD exclusion         |

## 4. AUTHORIZING OPHTHALMOLOGIST EVALUATION & SIGN-OFF

|                                                                                                                      |                                                                                          |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Authoring Doctor: Dr. S. Sharma, MD                                                                                  | Specialization: Senior Vitreo-Retina Specialist                                          |
| Medical License Reg No: MCI-RET-2026-889                                                                             | Hospital Affiliation: District Eye Hospital                                              |
| Clinical Status: Pending Review                                                                                      | Validation Timestamp: None                                                               |
| Doctor Prescriptions & Directives:<br>Verified presence of focal macular hard exudates. Routine follow-up scheduled. | Official Stamp & Digital Signature:<br><br>Signed Electronically by Authorized Clinician |

\* Generated by NetraAI Tele-Ophthalmology Network. Smart India Hackathon (SIH 2026). Stored permanently in MongoDB Atlas Cloud Cluster.